



JAD FALAQ

Data & AI Engineer

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6-month final-year engineering internship starting January 2027

PROFILE

Data & AI Engineer, final-year AI engineering student at **ENSIAS**, with experience at **Renault Digital** and **AKWA Group**. Specialized in **data pipelines**, **ERP integration**, **Machine Learning** and **generative AI**. Hybrid **Data Engineering / AI** profile, focused on **end-to-end automation**, **data governance** and **digital transformation of processes**.

Data Engineering – Machine Learning – GenAI – RAG – ETL

PROFESSIONAL EXPERIENCE

AI Engineer Intern – Quality Engineering

Renault Digital | Internship | Casablanca, Morocco

Jun. 2026 – Aug. 2026

- **AI Agents & Test Generation:** Designed and deployed to production a multi-agent LLM system (Requirement Analyzer Agent, Test Design Agent) turning requirements into Jira tickets and Gherkin scenarios, with human validation at each step (human-in-the-loop). POC on Copilot Studio and Power Platform, then full-stack rebuild (FastAPI, React/TypeScript). 100+ Jira tickets and 50 Gherkin scenarios generated, writing time cut from 10–20 min to 1–2 min per requirement, adopted by a 6-person QA team.
- **RAG & Document Ingestion:** Built a RAG on the Microsoft Copilot API (cosine similarity, top-5) fed by an ingestion pipeline of 400+ PDFs (Confluence, past user stories and Jira tickets): OCR of tables and images (unstructured.io), document-structure chunking and MMR retrieval on pgvector. DLP masking (Presidio, spaCy) before every LLM call.
- **Evaluation & Monitoring:** JSON generation validated by Pydantic schemas, then deterministic Gherkin rendering checked by the official Cucumber parser, eliminating LLM syntax errors. Similarity threshold calibration and quality dashboard, with 85% of generations accepted without edits.
- **QA Automation:** Generation of Smoke/regression test files on SharePoint and Teams reporting (Power Automate), deployed internally.

ML & Data Engineer – Data Team

AKWA Group | Internship | Casablanca, Morocco

Jun. 2025 – Sep. 2025

- **Data Engineering & ERP Integration:** Automated, with the Data team, the detection and ERP integration of missing transactions across 6M+ SQL records, gaining hands-on experience with large-scale data processing and real-world data quality issues.
- **Machine Learning & Anomaly Detection:** Built an end-to-end ML pipeline (preprocessing, feature engineering, modeling, evaluation) to detect and classify missing transactions.
- **Modeling & Evaluation:** Compared 5 algorithms (logistic regression, SVM, decision tree, Random Forest, KNN); the selected Random Forest reached 98% accuracy and 85% F1-score.

TECHNICAL SKILLS

- **Data Engineering & BI:** SQL, ETL, Pandas, Kafka, Power BI, document ingestion (unstructured.io, OCR), data quality (deduplication), ERP integration
- **Databases:** PostgreSQL, pgvector, MongoDB, Neo4j, SQLite / SQLAlchemy, Chroma
- **Generative AI & RAG:** Azure OpenAI, Microsoft Copilot API, LangChain, LangGraph, multi-agent systems, chunking, embeddings, hybrid search, MMR, RRF
- **Machine Learning & Deep Learning:** scikit-learn, PyTorch, Feature Engineering, YOLOv8, Transformer fine-tuning (AraBERT)
- **MLOps & Deployment:** MLflow, DVC, Docker, Git, GitHub Actions (CI/CD), Azure, Cloudflare
- **Back-end & Automation:** Python, FastAPI, Pydantic, REST APIs, Power Automate, Power Platform, SharePoint

EDUCATION

State Engineering Degree – Artificial Intelligence Engineering (2IA)

ENSIAS, Rabat | Sep. 2024 – Jun. 2027

- Machine & Deep Learning (neural networks, reinforcement learning), statistical and probabilistic learning
- Big Data and distributed systems, data engineering, time series, multimedia processing (text, image)

Preparatory Classes for the French Grandes Écoles (CPGE)

CPGE Mohammed V, Casablanca | Sep. 2022 – Jun. 2024

- Applied mathematics, engineering sciences and Python programming

ACADEMIC PROJECTS

- **Generative AI & RAG:** RAG chatbot on Azure OpenAI (gpt-4.1-mini, text-embedding-3-small) comparing several retrieval strategies (vector, hybrid, multi-query, reranking), with Simple and Agentic RAG modes and performance metrics displayed
- **Knowledge Graphs & GraphRAG:** Designed an explainable candidate-offer matching system on a Neo4j knowledge graph: skill weighting in Cypher, semantic retrieval and LLM explanations of fit and gaps (LangChain, FastAPI, Streamlit)
- **Computer Vision & MLOps:** Built a dental X-ray dataset (merged and cleaned several sources, train/validation/test split, YOLO annotations), detection of 7 pathologies with YOLOv8 (1,808 X-rays, 86% precision, DVC, MLflow)

SOFT SKILLS

- **Organization & Teamwork:** Logistics lead of the ENSIAS Forum Club (company–student meetings)
- **Autonomy & Problem Solving:** Delivered an AI solution to production under access and confidentiality constraints

LANGUAGES

French: Native – English: Bilingual – Spanish: B1